Search Engine Simulation Experiment

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> # Simulation Experiment for Search Engine Key Phrase Discovery
> set.seed(42) # Set seed for reproducibility
> # Test data probability
> p <- 0.60
> # Number of simulations
> N <- 10000
> # Simulate X: number of searches to the first success (Geometric distribution)
> # rgeom returns failures before the first success, so we add 1
> X <- rgeom(N, p) + 1
> # --- Task 1: Simulated PMF, Mean, and Variance ---
> mean_X <- mean(X)
> var_X <- var(X)
> cat("--- 1. Complete Set Statistics ---\n")

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> cat(sprintf("Simulated Mean: %.4f (Theoretical: %.4f)\n", mean_X, 1/p))

Simulated Mean: 1.6627 (Theoretical: 1.6667)

> cat(sprintf("Simulated Variance: %.4f (Theoretical: %.4f)\n", var_X, (1-p)/(p^2)))

Simulated Variance: 1.1294 (Theoretical: 1.1111)

> # Plot simulated PMF
> barplot(table(X) / N,
+         main = sprintf("Simulated PMF of X (p = %.2f)", p),
+         xlab = "Number of Searches (X)",
+         ylab = "Probability",
+         col = "steelblue",
+         border = "black")
> # --- Task 2: Conditional Distribution given X > 3 ---
> # Remaining searches after 3 failed attempts: X - 3 given X > 3
> X_gt_3 <- X[X > 3]
> X_remaining <- X_gt_3 - 3
> mean_cond <- mean(X_remaining)
> var_cond <- var(X_remaining)
> cat("\n--- 2. Conditional Distribution (X - 3 | X > 3) ---\n")

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> cat(sprintf("Conditional Mean: %.4f (Matches complete set mean: %.4f)\n", mean_cond, mean_

Conditional Mean: 1.6512 (Matches complete set mean: 1.6627)

> cat(sprintf("Conditional Variance: %.4f (Matches complete set variance: %.4f)\n", var_cond
```

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Conditional Variance: 1.1600 (Matches complete set variance: 1.1294)

> # --- Task 3: Markov Memoryless Property Estimates ---
> # Estimate P(X = 4 | X > 3) and P(X = 1)
> p_cond_4 <- sum(X == 4) / sum(X > 3)
> p_1 <- mean(X == 1)
> # Estimate P(X = 5 | X > 3) and P(X = 2)
> p_cond_5 <- sum(X == 5) / sum(X > 3)
> p_2 <- mean(X == 2)
> cat("\n--- 3. Memoryless Property Verification ---\n")

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> cat(sprintf("P(X = 4 | X > 3) = %.4f  |  P(X = 1) = %.4f\n", p_cond_4, p_1))

P(X = 4 | X > 3) = 0.5958  |  P(X = 1) = 0.6053

> cat(sprintf("P(X = 5 | X > 3) = %.4f  |  P(X = 2) = %.4f\n", p_cond_5, p_2))

P(X = 5 | X > 3) = 0.2695  |  P(X = 2) = 0.2370
```